SafePay: AI-Powered Fraud Detection System

Intelligent Transaction Monitoring & Fraud Risk Analysis



Project Team

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Academic Details

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Academic Project | [Guide Name] | [Date]

Problem & Objective

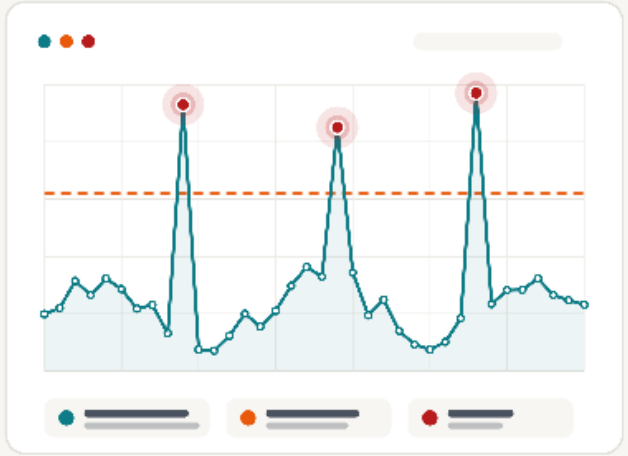
Problem Statement

Manual monitoring is slow and hard to scale.

- **Key fraud signals:** Very large amounts, rapid repeated transactions, and unusual behavior.

Key Objectives

- Detect suspicious transactions and score risk.
- Combine ML with explainable rules.
- Generate fraud alerts and keep an audit ledger.
- Support admin monitoring through a desktop application.



Normal Transaction → **Suspicious Activity** → **Financial Risk**

> *"Automating transaction assessment is vital to mitigating financial risk before fraud escalates at scale."*

Solution & Decision Outcomes

SafePay uses a multi-layer approach: validation, ML prediction, behavioral rules, limit and balance checks, security, persistence.

APPROVED

Valid and safe, balance updated.

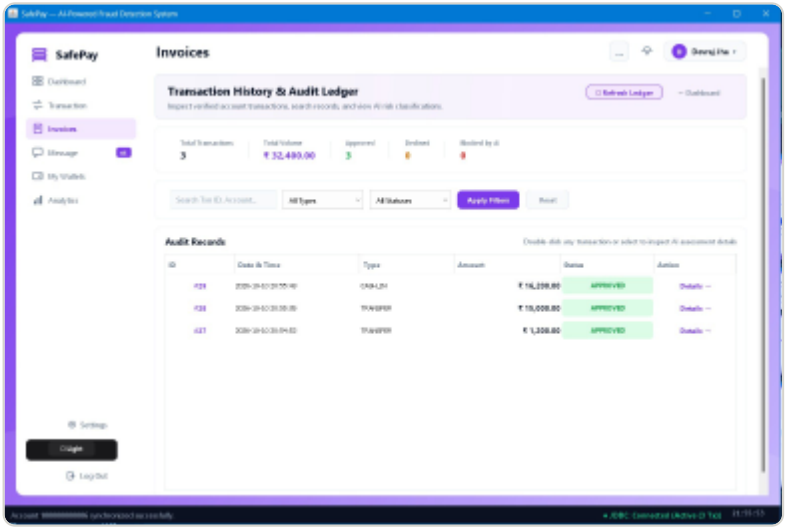
DECLINED

Valid but a business rule fails (limit or insufficient balance), not treated as fraud.

Limits per type: Transfer ₹200,000, Payment ₹100,000, Debit ₹100,000, Cash Out ₹50,000, Cash In ₹50,000.

BLOCKED

Suspicious, balance unchanged, alert raised.



Transaction History & Audit Ledger

System Workflow

Workflow for instant analysis when a transaction is submitted:

1. Submit Transaction
(Desktop Client > REST API)

2. Validate Input & Ownership > Load Account > Feature Extraction

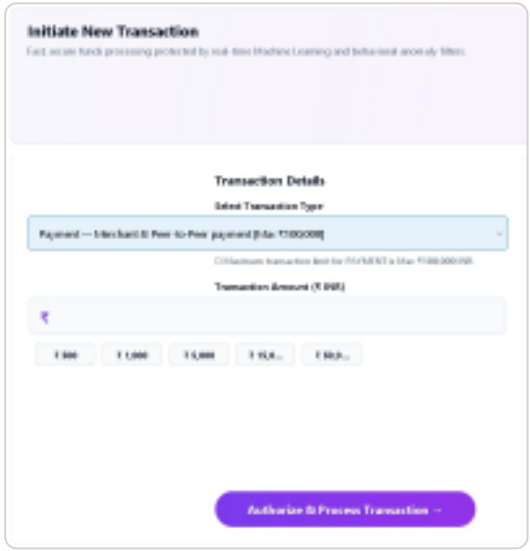
3. ML Prediction + Rule Checks

4. Suspicious? → BLOCKED

5. Limit exceeded? → DECLINED

6. Insufficient balance? → DECLINED

All outcomes are recorded in **MySQL** and shown in the client.



Step 1: user submits a transaction

Footnote: A blocked transaction never modifies the account balance.

Hybrid Detection: ML + Rules

Machine Learning Model (Weka / Random Forest)

- Learns patterns dynamically from historical transaction data.
- Outputs binary classification: **Legitimate** vs. **Fraud**.
- Calculates real-time fraud probability score.

Rule Engine (Behavioral Checks)

- Monitors behavioral velocity and anomaly thresholds.
- Flags multiple transactions in a short time window.
- Detects repeated large amounts combined with recent account activity spikes.

Hybrid Execution Detection Flow

ML Prediction
LEGITIMATE
Prob: 0.00%

→

Behavioral Analysis
SUSPICIOUS
Rule Triggered

→

Final Decision
BLOCKED
Action Enforced

→

Decision Reason
Behavioral Anomaly
Velocity Limit

Alert Severity & Risk Policy:
System alerts carry assigned severity levels (Medium, High/Critical). Any transaction blocked due to behavioral anomalies is automatically escalated and treated as **High Risk**.

System Architecture

User / Administrator

Java Swing Desktop App (+ Java2D charts)

REST API Client

Spring Boot REST API

Authentication	Fraud Detection	Dashboard Data
Spring Security	Rules + Weka ML	Analytics & Metrics

MySQL Database

"The desktop client communicates with the backend through HTTP API requests; the backend handles logic, security and data access."

Technology Stack

Desktop Client

- Java Swing
- Java2D (charts)
- REST API client

Backend

- Java 25
- Spring Boot REST API
- JDBC
- Maven Wrapper

Security

- Spring Security
- CSRF protection
- Transaction ownership checks

Database

- MySQL 8

Machine Learning

- Weka (Random Forest)

Testing & Tools

- JUnit
- Maven Surefire
- Git & GitHub

Machine Learning Pipeline

End-to-End Workflow Architecture



Core Execution Phases

01 Dataset Loading

02 Preprocessing

03 Feature Extraction

04 Training & Evaluation

Model Summary

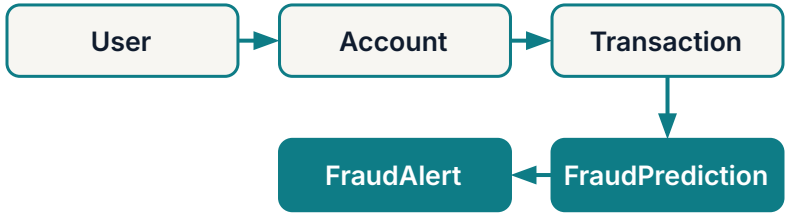
MODEL
Random Forest
Weka classifier trained on historical data

OUTPUT
Fraud Score
Legitimate vs. Fraud, scored in real time

DEPLOYMENT
ML + Rules
Combined with behavioral rule checks

Database & Security

Database ER Chain



"MySQL stores users, accounts, transactions, predictions and alerts."

Security Features

- Spring Security authentication & authorization
- Role-based access (USER / ADMIN)
- CSRF protection for API requests
- Transaction ownership checks
- Centralized exception handling and audit trail



Secure login: desktop client connected to the Spring Boot API



Role-based access: admin-only alert feed

User, Admin & Desktop Modules

User Module

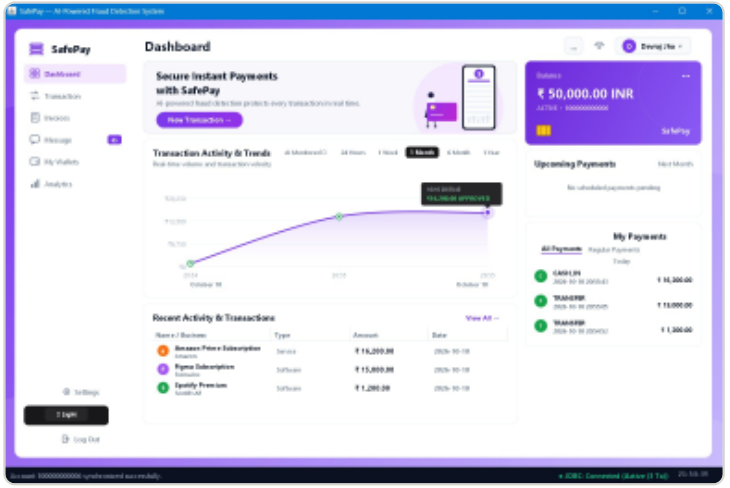
- Login/registration & transaction submission
- Dashboard with Java2D chart & transaction history

Admin Module

- Fraud alert feed with severity and status filters
- Strict ROLE_ADMIN-only access control

Desktop Client

- Swing screens with integrated Java2D charts
- Dedicated REST API client integration



SafePay Desktop Dashboard

Automated Test Coverage

Backend services, REST controllers, authentication and authorization, transaction processing, fraud rules, feature extraction, and desktop UI.

Scenarios & Future Scope

Transaction Scenarios

A) ₹500 Payment — APPROVED

ML legitimate, rules normal, all checks pass

B) ₹150,000 Payment — DECLINED

Exceeds ₹100,000 Payment limit

C) Large Transaction — BLOCKED

Large transaction after recent activity: ML legitimate, rules suspicious

Future Scope

- Desktop UI polish and Windows installer packaging
- More detailed fraud analytics
- Improved automated UI testing
- Real-time fraud-alert updates
- Expanded model evaluation and monitoring

Conclusion & Thank You

> *"SafePay combines a Java Swing desktop client, a secured Spring Boot REST API, rule-based checks and Weka machine learning to identify suspicious transactions and support administrative monitoring."*

Note: SafePay is a development project, not a certified payment-processing platform.

SYSTEM PIPELINE FLOW

Validate → Predict → Analyze → Decide → Alert → Monitor

Thank You | Questions & Discussion

Devraj Jha • Kriday Rajiv Kumar Singh • Ayush Rai • Ashutosh Maurya

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